



Research paper

Neurocognitive correlates of emotion-based urgency in DSM-5 bipolar disorder: the moderating role of diagnosis status and positive affect in a functional near-infrared spectroscopy (fNIRS) experimental study

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ABSTRACT

Background: A key goal of clinical neuroscience is to uncover reliable neurobiological targets for treatment. Urgency (negative or positive urgency subdomains, or N.urg/P.urg respectively), or affect-based impulsivity, is clinically relevant in bipolar disorder (BD). However, the neurocognitive correlates of positive and negative urgency have been mixed, with few considering the moderating effects of BD diagnosis and positive emotional context.

Methods: We conducted a 2 (group: BD vs. gender-matched healthy controls, or HC) by 2 (condition: positive vs. neutral affect) by 2 (task difficulty; within subjects) mixed, single-blind study. We examined neurocognitive correlates of urgency using EF performance and neuroimaging data from 74 participants (60 % females, $M_{age} = 36$).

Results: Between-group differences (HC > BD) in EF performance ranged from negligible to large (η_p^2 : 0 to 0.18). At the $p < .05$ threshold, positive urgency was correlated with lower cognitive control and working memory efficiency (r ranging from 0.25 to 0.34, $p < .05$). N.urg was correlated with reduced activity in the ventrolateral PFC (vLPFC) during cognitive control ($r = -0.26$ to -0.29 , $p < .05$). Multivariate Analysis of Variance (MANOVA) and subsequent simple slopes analyses revealed significant moderation by emotional context and BD diagnosis. High N.urg predicted hypoactivity in the left orbitofrontal cortex in BD only, whereas high P.urg predicted hyperactivity in the right vLPFC in positive emotion context condition only. No correlations/MANOVA findings remained significant after p-value correction.

Conclusion: Notwithstanding the study's limitations, the nature of the relationship between urgency and neurocognitive factors may be person- and context dependent.

1. Introduction

Urgency, defined as the tendency to engage in rash and regrettable actions triggered by intense emotional states, has been shown to be relevant to psychopathology (Berg et al., 2015), maladaptive decision-making (Cyders et al., 2016; Cyders and Smith, 2008; Cyders et al., 2010; Elliott et al., 2023b), substance misuse (Coskunpinar et al., 2013; Hershberger et al., 2017; Kale et al., 2018; Um et al., 2019), suicidality

(Bruno et al., 2023), and poorer treatment response (Hershberger et al., 2017; Peckham et al., 2019). Urgency has also been shown to have robust predictive validity for the worsening and onset of many different clinical conditions, including non-suicidal self-harm, alcohol and drug use (Bruno et al., 2023; Gunn et al., 2020; Ortin-Peralta et al., 2024; Riley et al., 2015).

Urgency has major clinical significance in bipolar disorder (BD), being associated with mania risk, trauma, aggression, functional

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impairment, and reduced quality of life in BD (Giovannelli et al., 2013; Muhtadie et al., 2014; Teh et al., 2024; Victor et al., 2011). When compared to other non-emotion-based impulsivity subdimensions, urgency has been shown to be more strongly associated with bipolar diagnoses (Muhtadie et al., 2014; Teh et al., 2023).

Investigating the neurocognitive mechanisms of urgency is crucial, aligning with current efforts to elucidate brain mechanisms involved in neurocognitive, social, and emotional processes, as outlined by the Research Domain Criteria (RDoC) framework (Cuthbert, 2022). This approach aims to identify new neurobiological targets in BD (Miskowiak et al., 2017; Miskowiak et al., 2019), which could facilitate refining existing assessments and interventions by incorporating novel, clinically relevant psychosocial factors into the broader biopsychosocial framework (Bolton, 2023).

Dominant neurocognitive theory suggests that urgency is explained, in part, by suboptimal self-control and decision making (Carver et al., 2008; Dekker and Johnson, 2018; Johnson et al., 2016; Smith and Cyders, 2016). Contrary to theory, the neural correlates of urgency have been highly varied, with some evidence indicating weaker resting-state functional connectivity and morphometry involving the orbitofrontal cortex (OFC; Elliott et al., 2023a; Johnson et al., 2020). Possible reasons for those inconsistencies have been well documented (Tiego et al., 2023); one key factor is the lack of experimental validity, in that studies using functional MRIs often lack a mood induction component. As urgency is an affect driven construct of impulsivity, an experimental manipulation of emotion is required to elicit a neural profile that correlates with its defined construct (Elliott et al., 2023b; Um et al., 2019).

Beyond building a better understanding of urgency, a core goal in this study was to examine how neurocognitive facets of urgency may be moderated by BD diagnosis. Past studies show that BD is associated with poorer executive function (EF) performance, and reduced neural activation in the dorsolateral (dlPFC), dorsomedial (dmPFC), ventral PFCs (vlPFC), among other frontoparietal regions during cognitive tasks with emotional stimuli (Ellard et al., 2019; Phillips and Swartz, 2014; Yamamuro et al., 2018). Only a few studies have considered how urgency might relate to neurocognitive correlates of BD and (hypo)mania (Johnson et al., 2019; Schumer et al., 2022). Thus, this study aimed to elucidate 1) neurocognitive correlates of urgency, and 2) the moderators of the relationship between urgency and neurocognitive executive function (EF) outcomes. Drawing from research described above, we hypothesized that 1) BD would perform more poorly than healthy controls (HC) on EF measures and show reduced haemodynamic activation (main effects of group); 2) urgency (positive or negative subdimensions) will predict EF outcomes and PFC cortical activity (main effects of urgency), and lastly we explored three-way interactions (i.e., urgency $\times$ group $\times$ condition) to further explicate the complex relationships between urgency, BD, and positive emotion. These hypotheses have been preregistered (<https://osf.io/rquqe>).

2. Method

This study used a mixed 2 (BD vs HC; between subjects) by 2 (positive vs neutral emotion induction condition; between subjects) by 2 (level task difficulty; within subjects) experimental design. The outcome variables were behavioural performance and neural activity during cognitive control and working memory tasks, which are two important EF abilities (Friedman and Robbins, 2022). Power analysis was conducted a priori (see supplementary material). This study received ethics approval from the Institutional Review Board (IRB) of the Institute of Mental Health (IMH) and the Domain Specific Review Board (DSRB) of the National Health Group (NHG) of Singapore (DSRB No. 2022/00609). All participants gave written informed consent before their participation, in the presence of a witness.

2.1.1. Materials & assessments

Demographic questionnaire: a pen-and-paper questionnaire of age, biological sex, highest education attained, hand dominance, caffeine intake in the past 6 h (yes/no), and hours of sleep. Medications were extracted from medical records.

Short Impulsivity Behavior Scale (S-UPPS-P): this 20-item scale includes five behavioural impulsivity subscales: Negative and positive urgency, lack of perseverance, premeditation, and sensation-seeking. Responses are rated on a 4-point Likert scale from strongly disagree to strongly agree (Cyders et al., 2014). This scale has been validated in a psychiatric setting (Dugré et al., 2019). The internal reliabilities for subfactors were 0.80 (negative urgency), 0.79 (positive urgency), 0.51 (lack of perseverance), 0.8 (lack of premeditation), 0.75 (sensation seeking).

The Structural Clinical Interview for Diagnostic and Statistical Manual of Mental Disorders, 5th edition or DSM-5 (SCID-5) was used to screen for the inclusion and exclusion criteria for the study (elaborated in the supplementary material).

Montgomery-Asberg Depression Rating Scale (MADRS) and Young Mania Rating Scale (YMRS) were used to assess depression and manic severity in BD participants.

Visual Analogue Scale (VAS): Participants used a digital 10 cm line to rate how happy or sad they felt at various points during the experiment (pre-induction, post-induction, after first EF task, after second EF task at the end of the study). Scores range from -50 (sad) to 50 (happy).

2.1.2. Cognitive tasks

We chose two of the most used measures of executive function, the N-back and the Stroop colour-word task.

N-back task for working memory: the n-back task is a test of working memory capacity (Frost et al., 2021). Participants completed 1-back and 2-back trials, with the latter representing greater cognitive load. We recorded the changes in the number of hits (correct responses), errors (responding to non-targets), and reaction times taken between 2-back and 1-back trials. Larger increases in errors and reaction times (or larger decreases in hits) indicate greater difficulty in managing the heightened cognitive load associated with 2-back demands.

A modified Stroop colour-word task (SCWT): The modified SCWT assesses an individual's ability to inhibit prepotent automatic responses (a form of cognitive control; Friedman and Robbins, 2022). Participants were presented alternating blocks of trials with words "RED", "BLUE", or "GREEN" in the black colour font, termed word condition (or congruent trial; W), and colour words in red, blue, or green ink, such that the font colour never matched the colour word, termed colour word condition (or incongruent trial; CW). We recorded changes in frequency of hits (correct responses), errors (commission responses), and average reaction time (correct responses only), and calculated the inverse efficiency score (IES) by dividing the average reaction time (of correct responses) with the percentage of hits (Yeung et al., 2020). In addition, lower IES values indicate better processing efficiency. We focused on changes in error scores and reaction time, which serve as indicators of Stroop interference (cognitive control).

2.1.3. Functional near-infrared spectroscopy (fNIRS) equipment

A non-invasive, head-mounted 48-channel near-infrared spectroscopy (NIRS) system (NIRSIT, OBELAB, South Korea) was used to measure cortical activity through blood oxygenation changes. NIRS signals have shown to significantly correlate with BOLD signals at the group level (Sato et al., 2013; Wijekumar et al., 2017). The modified Beer-Lambert Law was used to convert raw light intensities into changes in oxygenated haemoglobin (ΔHbO_2). Average ΔHbO_2 was calculated by pre-processing raw fNIRS signals (see supplementary material) into channel-wise beta estimates for 8 regions of interest (ROI): left and right

dorsolateral prefrontal cortex (dlPFC), ventrolateral prefrontal cortex (vlPFC), medial prefrontal cortex (mPFC), and orbitofrontal cortex (OFC), which were categorized by the 48 channels (See supplementary section for an elaboration).

2.1.4. Participants

Participants were 1) patients receiving outpatient treatment for bipolar I disorder (BD) from the Institute of Mental Health (IMH), a tertiary psychiatric hospital in Singapore, who were assessed for DSM-5 diagnosis of bipolar I, using the SCID-5; and 2) HC were recruited from the community by word-of-mouth. A total of 148 individuals were approached but 42 declined to participate and 26 did not meet study criteria. A total of 80 individuals completed the study (6 of whom were part of a pilot test, and therefore not reported), thus leaving a total of 74 individuals' data (BD $n = 37$, HC $n = 37$) extracted. The response rate was 54.1 % (78.7 % for HC, 42.6 % for BD). See the supplementary material for full inclusion and exclusion criteria.

2.1.5. Procedure

On the testing day, BD participants first completed MADRS and YMRS interviews, while HC did not. Next, all participants provided sociodemographic information and completed the S-UPPS-P survey. Once the fNIRS was successfully fitted and calibrated, the principal investigator (PI) proceeded to read instructions verbatim. Participants proceeded to perform the tasks on a laptop in a well-lit and quiet room, supervised by the PI behind a one-way mirror. Participants completed the first VAS at pre-induction, then the modified Go task. Participants randomized into the positive condition received a bogus feedback stating their responses on the Go task were "very fast," while the neutral group received no feedback. Afterward, they rated their mood again (post-induction VAS). Participants completed the SCWT and n-back tasks in counter-balanced order and completed a VAS after each task (denoted as T1 and T2; see supplementary material Fig. 1). At the end of the experiment, participants were debriefed and reimbursed SGD \$80 dollars.

2.1.6. Analysis plan

The outcome variables were behavioural performance and neural activity during cognitive control and working memory tasks, which are two important EF abilities (Friedman and Robbins, 2022). Power analysis was conducted a priori. Raw optical signals were pre-processed, and missing values were handled with mean imputation (see supplementary material).

2.1.6.1. T-tests, correlations, univariate, multivariate analyses. t-tests, correlational analyses, and repeated measures mixed ANOVAs were conducted to assess demographic characteristics and BD vs. HC group differences. To examine the effects of the positive emotion manipulation, two separate univariate repeated measures mixed analysis of variance (ANOVA) were performed to evaluate any interaction or main effect of group by experimental condition (neutral or positive condition) on 1) VAS scores, and 2) PFC haemoglobin change (ΔHbO_2). The results are reported in the supplementary section. Bonferroni adjustments were applied to account for multiple tests.

The Bonferroni adjustment method is conservative and may lead to overcorrection, increasing the risk of Type II errors wherein true effects are erroneously dismissed as non-significant (Hochberg, 1988). Applying Bonferroni adjustments to the current small sample would reduce its power to detect true results. To balance out these concerns of type I and II errors, we reported results based on the two p value thresholds – Bonferroni adjusted and standard $p < .05$ thresholds.

To examine the effects of urgency, group (BD vs. HC), and condition (neutral vs. positive emotional context) on EF and cortical activity outcomes separately, a total of 8 multivariate analysis of variance (MANOVA within the GLM framework) were conducted to examine the

effects of 2 (neural activity/cognitive scores) $\times$ 2 (separate models for positive/negative urgency) $\times$ 2 (cognitive control/working memory) on the neurocognitive variables, while controlling for sleep duration. Bonferroni adjusted threshold ($p < .006$) and the conventional ($p < .05$) cut-off were used.

Separate, parallel models of positive and negative urgency were used 1) due to insufficient sample size for the analysis of four-way interactions (e.g., group $\times$ condition $\times$ negative urgency $\times$ positive urgency), 2) because positive and negative urgency have been shown to be associated with differential clinical outcomes despite being moderately correlated (Teh et al., 2023), and 3) the correlation between the two subdomains found in this current investigation are high ($r = 0.77$). For MANOVA, we first considered whether the three-way interactions were significant, and if the terms were not significant, then simpler models (i.e., two-way or main effects model) were presented. We proceeded to conduct multiple regression analyses to ascertain the location of the multivariate effect. Any significant interactions were further probed with post hoc simple slopes tests.

Descriptive statistics, t -tests, correlation analyses, repeated measures were conducted on IBM SPSS version 23, whereas MANOVA and multivariate analysis of covariance (MANCOVA) analyses were conducted in RStudio, version 2024.04.0+ 735. Multiple linear regression analyses were performed post-hoc to locate the outcomes that showed significant effects.

2.1.6.2. Transparency and openness. The analysis plan differs from the pre-registered version. The change was ultimately due to concerns over convergence issues in Linear Mixed Model (LMM), arising from the small sample size relative to the number of variables, as well as the updated knowledge of the need for separate models for negative and positive urgency, which ultimately led us to adopt MANOVA as a more suitable approach after pre-registering.

In addition, due to the low correlations observed between all 8 PFC regions, cognitive control, and working memory performance indicators, 8 parallel multivariate analysis of variance (MANOVA) analyses were conducted separately for cognitive performance and cortical activity (ΔHbO_2 data), Stroop and N-back tasks, and positive and negative urgency.

3. Results

3.1. Sample characteristics

The sample's characteristics are summarised in Table 1. Participants were 36.01 years old on average ($SD = 10.58$). The sample comprised of ethnic Chinese (73 %), Malay (15 %), Indian (8 %), and other ethnic groups (4 %). Most had university or higher education (55.4 %) and were right-handed (81.1 %). Participants reported 7.6 ($SD = 1.51$) hours of sleep on average. BD and HC groups were matched by biological sex. A small percentage of participants had caffeine in the past 6 h (13.5 %). There were no significant group-differences in age, sex (matched), educational attainment, hand dominance, or caffeine consumption between BD & HC groups. However, BD ($N = 37$; Mean = 8.07 h; Standard Deviation (SD) = 1.81) reported significantly greater duration of sleep than HC ($N = 37$, $M = 7.16$ h; $SD = 0.97$), $t(51.1) = -2.71$, $p = .008$, which was included as a covariate in all MANOVA and MANCOVA analyses.

Regarding medications, most patient participants were prescribed at least 1 mood stabilizer (78.4 %) and at least 1 antipsychotic medication (83.8 %). In contrast, only a minority of patient participants were prescribed at least 1 antidepressant medication (13.5 %), and less than half of patient participants were prescribed anxiolytics or sedatives (45.9 %).

Table 1

Descriptive summary of Bipolar I (n = 37) and Healthy Controls (n = 37) groups.

		Total (n = 74) M(SD) / %	BD I (n = 37) M(SD) / %	HC (n = 37) M(SD) / %	t / χ^2	df	p	Positive (n = 38) M(SD) / %	Neutral (n = 36) M(SD) / %	t / χ^2	df	p
1	Age	36.01 (10.58)	35.95 (10.64)	36.08 (10.7)	0.05	71.9	0.96	37.55 (11.82)	34.39 (8.97)	-1.30	68.8	0.2
2	Sex									0.44	1	0.5
	Male	40.50 %	40.50 %	40.50 %	matched			36.80 %	44.40 %			
	Female	59.50 %	59.50 %	59.50 %				63.20 %	55.60 %			
3	Highest Education				1.37	2	0.50			0.54	2	0.8
	Secondary & lower	9.50 %	10.80 %	8.10 %				5.30 %	8.30 %			
	Pre-university	35.10 %	40.60 %	29.70 %				34.20 %	27.80 %			
	University & above	55.40 %	48.60 %	62.20 %				60.50 %	63.90 %			
4	Hand dominance				0.79	1	0.37			0.50	1	0.5
	Right	81.10 %	86.50 %	75.70 %				77.80 %	22.20 %			
	Left	18.90 %	13.50 %	24.30 %				84.20 %	15.80 %			
	Sleep (hrs)	7.60 (1.51)	8.07 (1.81)	7.16 (0.97)	-2.71	55.1	0.008	7.60 (1.59)	7.64 (1.44)	0.11	72	0.9
5	Caffeine taken									0.20	1	0.7
	Yes	13.50 %	89.20 %	83.80 %				10.50 %	13.90 %			
	No	86.50 %	10.80 %	16.20 %				89.50 %	86.10 %			
6	VAS	18.24 (21.12)	13.38 (22.13)	23.11 (19.13)	2.02	72	0.05	19.82 (21.80)	16.58 (20.54)	-0.66	72	0.5
7	Counter-balance									1.41	1	0.2
	Stroop first	54.10 %	54.10 %	54.10 %	Matched			47.40 %	61.10 %			
	N-back first	45.90 %	45.90 %	45.90 %				52.60 %	38.90 %			

		BD I (n = 37) M(SD) / %	Positive (n = 19) M(SD) / %	Neutral (n = 18) M(SD) / %	t / χ^2	df	p
1	Mood stabilizers				0.62	2	0.7
	0 prescribed	21.60 %	26 %	17 %			
	1 prescribed	59.50 %	58 %	61 %			
	2 prescribed	18.90 %	16 %	22 %			
2	Antipsychotics				0.69	3	0.9
	0 prescribed	16.20 %	21.10 %	11.10 %			
	1 prescribed	51.40 %	47.40 %	55.60 %			
	2 prescribed	21.60 %	21.10 %	22.20 %			
	3 prescribed	10.80 %	10.50 %	11.11 %			
3	Antidepressants				2.1	2	0.4
	0 prescribed	86.50 %	78.90 %	94.40 %			
	1 prescribed	10.80 %	15.80 %	5.60 %			
	2 prescribed	2.70 %	5.30 %	0.00 %			
4	Anxiolytics/Sedatives				1.38	3	0.7
	0 prescribed	54.10 %	52.60 %	55.60 %			
	1 prescribed	35.10 %	31.60 %	38.90 %			
	2 prescribed	8.10 %	10.50 %	5.56 %			
	3 prescribed	2.70 %	5.30 %	0.0 %			
5	MADRS	5.68 (6.66)	5.16 (6.44)	6.59 (6.86)	0.912	70	0.4
6	YMRS	2.43 (3.38)	2.32 (3.74)	2.47 (3.02)	0.192	70	0.8

Items 1–6 were self-reported; items 7 & 8 were interviewer-administered scales conducted before the experiment; item 9 is the Visual Analogue Scale that was measured in between experimental tasks; items 10–12 were medications prescribed, extracted from medical records; BD I is bipolar I; HC is healthy control; M is mean value; SD is standard deviation; df is degrees of freedom; p value is significant at <0.05; MADRS is Montgomery-Asberg Depression Rating Scale; YMRS is Young Mania Rating Scale; SUPPS-P is the short impulsivity behavioural scale;

3.2. Positive mood induction

The experimental manipulation, while mild in intensity, had a significant influence on participants at the neural level (ΔHbO_2) but not at the subjective level (VAS scores). Results indicated a significant group $\times$ condition interaction [$F(1,70) = 7.56, p = .008$] and a significant main effect of condition [$F(1,70) = 5.07, p = .027$] on ΔHbO_2 activation. Further details are reported in the supplementary section.

3.3. Stroop, N-back performances, and urgency differences between groups

Refer to Table 2 for the full summary of results. At the Bonferroni-adjusted p threshold, BD ($M = 2.34, SD = 0.57$) scored significantly higher on positive urgency than HC ($M = 1.82, SD = 0.53$). This difference was large (Cohen's $d = 0.95$). No significant differences were observed for negative urgency, and other non-emotion based impulsivity subdomains.

BD recorded significantly fewer correct responses (or hits; HC – BD $M_{\text{diff}}: 15.8$), and slower reaction time ($M_{\text{diff}} = -0.29$) than HC on Stroop, and slower reaction time than HC on the N-back task. At the unadjusted p threshold, BD scored significantly fewer number of hits than HC. The groups did not differ significantly on the number of errors made during either task.

3.4. Correlations between urgency, EF performance, PFC cortical activity changes

At the Bonferroni-adjusted $p < 0.05$ cut-off, all positive and negative urgency subdomains were not significantly correlated with behavioural (Table 3) and neural outcomes (Table 4).

However, at the unadjusted threshold of $p < .05$, few correlations were significant. For Stroop, positive urgency was significantly correlated with slower reaction time in both W and CW conditions, fewer hits in the CW condition, and poorer efficiency (IES) in both W and CW conditions. Negative urgency was significantly and inversely correlated

Table 2

Stroop & N-back outcomes between BD and HC groups.

1	SUPPS-P Urgency	HC (n = 37) M(SD)	BD (n = 37) M(SD)	Total (n = 74) M(SD)	t	df	p ^a	d
	Emotional urgency	3.87 (0.93)	4.60 (1.10)	4.23 (1.07)				
	Positive urgency	1.82 (0.53)	2.34 (0.57)	2.08 (0.61)	-4.04	71.58	<0.001*	0.95
	Negative urgency	2.05 (0.54)	2.26 (0.60)	2.16 (0.58)	-1.63	71.02	0.11	0.37
	Premeditation (lack)	1.87 (0.39)	2.00 (0.51)	1.93 (0.45)	-1.29	67.49	0.20	0.29
	Perseverance (lack)	1.95 (0.42)	2.12 (0.48)	2.04 (0.45)	-1.62	70.54	0.11	0.38
	Sensation seeking	2.41 (0.60)	2.41 (0.70)	2.41 (0.65)	0.05	70.19	0.96	0.65
2	VAS (perceived mood)	M(SD)	M(SD)	M(SD)	t	df	p ^a	d
	Pre-induction	23.11 (19.13)	13.38 (22.13)	18.24 (21.12)	2.02	70.53	0.05	-0.47
	Post-induction	24.32 (16.79)	15.59 (19.05)	19.96 (18.37)	2.09	70.88	0.04	-0.49
	After 1st task	18.97 (17.12)	9.64 (22.88)	14.37 (20.57)	1.97	64.82	0.05	-0.46
	End 2nd task	20.65 (18.66)	14.78 (21.05)	17.72 (19.97)	1.27	70.97	0.21	-0.29
3	Stroop (cognitive control)	HC (n = 36) M(SD)	BD (n = 36) M(SD)	Mean diff HC - BD	F	df	p ^b	η_p^2
	Hits (W)	113.86(14.23)	97.78(20.1)	15.77	16.13	1	<0.001*	0.18
	Hits (CW)	98.05(19.09)	82.59(17.4)					
	Errors (W)†	3.97(13.09)	4.03(15.82)	-0.115	0.003	1	0.960	0.00
	Errors (CW)†	3.73(3.88)	3.92(9.06)					
	RT (W)	0.62(0.17)	0.88(0.37)	-0.286	14.84	1	<0.001*	0.17
	RT (CW)	0.8(0.36)	1.11(0.41)					
4	N-back (working memory)	HC (n = 36) M(SD)	BD (n = 36) M(SD)	Mean diff HC - BD I	F	df	p ^b	η_p^2
	Hits (1 N)	7.62(0.59)	7.19(1.06)	0.664	5.684	1	0.02	0.07
	Hits (2 N)	10.57(1.97)	9.67(1.88)					
	Errors (1 N)	0.19(0.4)	0.58(0.73)	-0.223	2.645	1	0.11	0.04
	Errors (2 N)	0.89(1.08)	0.94(0.89)					
	RT (1 N)	0.48(0.13)	0.53(0.19)	-0.115	10.924	1	0.001*	0.13
	RT (2 N)	0.48(0.22)	0.65(0.18)					

Note: rows labelled 1 (SUPPS-P) and 2 (VAS) are based on t-tests; 3 (Stroop) & 4 (N-back) are based on repeated measures mixed ANOVA; W is word condition; VAS is visual analogue scale; CW is colour word condition; RT is reaction time; M is mean; SD is standard deviation; HC is control group; BD is bipolar group; t is t-test statistic; df is degrees of freedom; diff the mean difference; d is Cohen's d; η_p^2 is partial eta-squared; Bonferroni adjusted p^a < 0.0055 (for 9 t-tests); Bonferroni adjusted p^b < 0.008 * (for 6 repeated mixed ANOVA); underlined p values are significant at unadjusted p < .05;

Table 3

Correlations between positive, negative urgency, and behavioural outcomes of Stroop and N-back tasks.

Stroop (n = 74)		Word (W) condition			Colour word (CW) condition			IES			
		Hits	Errors†	RT	Hits	Errors†	RT	IES (W)	IES (CW)†		
PU	Positive urgency	−0.21	−0.16	<u>0.29</u>	<u>−0.23</u>	−0.01	<u>0.28</u>	<u>0.25</u>	<u>0.34</u>		
NU	Negative urgency	−0.05	0.03	0.13	−0.11	0.19	0.17	0.16	0.16		
PER	Perseverance (lack)	0.04	−0.05	−0.06	0.13	<u>−0.28</u>	−0.11	−0.15	−0.07		
PM	Premeditation (lack)	−0.15	−0.11	0.14	−0.12	−0.14	0.14	0.12	0.17		
SS	Sensation seeking	0.15	0.11	−0.11	0.11	0.19	−0.12	−0.18	−0.14		
N-BACK (n = 73)		1-back				2-back				IES	
		Hits†	Errors	Misses†	RT†	Hits†	Errors	Misses†	RT†	1-back†	2-back†
PU	Positive urgency	−0.12	0.12	0.12	<u>0.28</u>	<u>−0.23</u>	−0.03	<u>0.23</u>	<u>0.24</u>	<u>0.34</u>	<u>0.25</u>
NU	Negative urgency	−0.02	0.10	0.02	<u>0.11</u>	−0.06	0.01	0.06	0.16	0.14	0.11
PER	Perseverance (lack)	−0.06	−0.01	0.06	0.04	0.06	0.07	−0.06	0.06	0.06	0.02
PM	Premeditation (lack)	0.03	0.04	−0.03	0.17	0.07	0.02	−0.07	0.03	0.17	0.01
SS	Sensation seeking	−0.05	0.16	0.05	−0.09	0.15	0.01	−0.15	0.01	−0.03	−0.12

Note: RT is reaction time; IES is inverse efficiency score which is the ratio of reaction time (RT) to the percentage of hits, higher values denote poorer efficiency; PU, NU, PER, PM, SS are subdomains of the UPPS-P impulsivity scale; † denote non-normal distribution, Spearman's rho was conducted; Bonferroni adjusted p < 5.55e⁻⁴ *(for 90 correlational analyses); underlined unadjusted p < .05.

Table 4Correlations between positive, negative urgency, and oxy-haemodynamic changes (ΔHbO_2) in the prefrontal cortex.

During stroop (n = 72)		dlpfc		fpc		ofc		vlpfc	
		Left†	Right†	Left†	Right	Left	Right†	Left†	Right†
PU	Positive urgency	0.04	0.03	−0.18	−0.06	−0.15	−0.15	−0.08	−0.18
NU	Negative urgency	−0.04	0.02	−0.10	0.04	−0.16	−0.14	−0.29	−0.26
PER	Perseverance (lack)	−0.02	−0.01	0.10	0.07	−0.03	−0.04	0.02	−0.13
PM	Premeditation (lack)	0.09	−0.12	−0.16	0.12	−0.04	0.08	0.02	0.01
SS	Sensation seeking	0.07	−0.02	−0.22	0.09	0.10	<u>0.24</u>	0.12	0.01
During N-BACK (n = 72)		dlpfc		fpc		ofc		vlpfc	
		Left	Right	Left†	Right	Left	Right†	Left	Right†
PU	Positive urgency	0.04	−0.13	0.20	0.10	0.01	0.08	0.10	−0.06
NU	Negative urgency	0.00	−0.07	0.15	0.10	−0.07	−0.05	−0.03	−0.02
PER	Perseverance (lack)	0.02	−0.21	0.04	−0.02	0.02	0.14	−0.01	−0.08
PM	Premeditation (lack)	−0.14	−0.09	<u>0.34</u>	0.04	0.09	0.19	−0.06	0.01
SS	Sensation seeking	−0.10	−0.04	−0.06	−0.21	<u>−0.25</u>	0.09	<u>−0.24</u>	0.00
During bogus positive feedback (positive condition only, n = 38)		dlpfc		fpc		ofc		vlpfc	
		Left	Right	Left†	Right	Left	Right	Left	Right
PU	Positive urgency	<u>0.37</u>	0.19	0.00	0.15	<u>0.34</u>	<u>0.36</u>	0.44	0.26
NU	Negative urgency	<u>0.43</u>	<u>0.33</u>	0.13	0.12	0.28	<u>0.41</u>	0.40	0.28
PER	Perseverance (lack)	−0.12	0.19	−0.06	0.26	0.11	−0.01	−0.04	0.01
PM	Premeditation (lack)	−0.13	−0.15	0.05	0.16	0.16	0.10	0.12	0.05
SS	Sensation seeking	−0.10	−0.08	0.15	−0.26	−0.03	0.19	−0.02	0.01
After bogus positive feedback (positive condition only, n = 38)		dlpfc		fpc		ofc		vlpfc	
		left	right	left	right	left†	right	left	right
PU	Positive urgency	0.18	0.20	−0.11	−0.03	0.07	0.10	0.24	0.24
NU	Negative urgency	0.26	0.29	−0.03	−0.06	0.04	0.19	0.19	0.28
PER	Perseverance (lack)	−0.19	0.00	0.05	0.10	−0.01	0.06	−0.10	−0.05
PM	Premeditation (lack)	−0.12	−0.19	−0.03	−0.05	−0.07	0.16	0.04	0.14
SS	Sensation seeking	−0.08	0.00	0.03	<u>−0.42</u>	−0.13	0.03	<u>0.33</u>	0.18
Corresponding screen without positive feedback (Neutral condition only, n = 36)		dlpfc		fpc		ofc		vlpfc	
		Left	Right	Left†	Right	Left	Right	Left	Right
PU	Positive urgency	−0.21	0.06	0.20	0.03	0.18	0.30	−0.12	−0.06
NU	Negative urgency	−0.23	−0.04	0.00	0.02	−0.10	0.20	−0.17	0.05
PER	Perseverance (lack)	0.04	0.15	−0.12	−0.23	0.16	0.13	0.08	−0.11
PM	Premeditation (lack)	−0.07	0.06	0.12	−0.16	−0.17	−0.04	0.02	0.02
SS	Sensation seeking	−0.04	−0.17	0.11	−0.10	−0.12	<u>−0.34</u>	−0.19	−0.16
Post-induction screen without positive feedback (neutral condition only, n = 36)		dlpfc		fpc		ofc		vlpfc	
		Left	Right	Left	Right	Left†	Right	Left	Right
PU	Positive urgency	−0.01	0.04	−0.07	−0.10	0.27	0.22	−0.05	−0.02
NU	Negative urgency	−0.17	−0.21	−0.17	−0.10	−0.11	0.07	0.16	−0.05
PER	Perseverance (lack)	0.09	0.14	<u>−0.34</u>	<u>−0.48</u>	−0.14	−0.03	0.01	−0.11
PM	Premeditation (lack)	−0.03	−0.05	−0.13	−0.24	−0.11	−0.10	0.13	0.05
SS	Sensation seeking	−0.04	−0.07	<u>0.33</u>	0.12	−0.09	−0.25	−0.18	−0.27

Note: PU, NU, PER, PM, SS are subdomains of the S-UPPS-P impulsivity scale; † denote non-normal distribution & results are based on Spearman's; Bonferroni adjusted $p < 2.08e^{-4}$ * (for 240 correlation analyses); underlined values $p < .05$.

with left ($r = -0.29$) and right ($r = -0.26$) vLPFC activity (Table 4). For the N-back, positive urgency was significantly correlated with slower reaction time, greater task inefficiency in both 1-back and 2-back conditions, and fewer number of hits in the 2-back condition. Negative urgency was not significantly correlated with all behavioural outcomes, and positive urgency was not significantly correlated with all ΔHbO_2 outcomes.

3.5. MANOVA¹

All interaction and main effects were not significant at the Bonferroni adjusted $p < .006$ threshold for all 8 models. See Table 5 for the

¹ Results were similar when sleep duration was not controlled for in the analyses.

Table 5
MANOVA summary.

Negative urgency model																
Three-way interaction	Stroop								N-back							
	Cortical activity (ΔHbO_2)				Behavioural outcomes				Cortical activity (ΔHbO_2)				Behavioural outcomes			
	Pillai's Trace	F	df	p	Pillai's Trace	F	df	p	Pillai's Trace	F	df	p	Pillai's Trace	F	df	p
Group $\times$ condition $\times$ N.urgency	0.09	0.65	(8, 56)	0.73	0.02	0.37	(3, 63)	0.77	0.14	1.11	(8, 56)	0.37	0.00	0.03	(3, 63)	0.99
Two-way interaction																
Group $\times$ condition	0.25	2.33	(8, 57)	<u>0.030</u>	0.03	0.62	(3, 64)	0.60	0.21	1.86	(8, 57)	0.08	0.10	2.24	(3, 64)	0.09
Group $\times$ N.urgency	0.28	2.78	(8, 57)	<u>0.010</u>	0.04	1.00	(3, 64)	0.40	0.15	1.25	(8, 57)	0.29	0.08	1.83	(3, 64)	0.15
Condition $\times$ N.urgency	0.07	0.53	(8, 57)	0.83	0.01	0.30	(3, 64)	0.82	0.13	1.09	(8, 57)	0.38	0.11	2.53	(3, 64)	0.07
Main effects																
Group	0.3 [^]	3.04 [^]	(8, 57)	<u>0.006</u>	0.09	2.33	(3, 67)	0.08	0.12	1.02	(8, 60)	0.43	0.13	3.32	(3, 67)	<u>0.03</u>
Condition	0.05 [^]	0.39 [^]	(8, 57)	0.92	0.01	0.23	(3, 67)	0.88	0.09	0.73	(8, 60)	0.66	0.01	0.24	(3, 67)	0.87
N.urgency	0.15 [^]	1.23 [^]	(8, 57)	0.30	0.02	0.36	(3, 67)	0.78	0.11	0.92	(8, 60)	0.51	0.00	0.08	(3, 67)	0.97
Positive urgency model																
Three-way interaction	Stroop								N-back							
	Cortical activity (ΔHbO_2)				Behavioural outcomes				Cortical activity (ΔHbO_2)				Behavioural outcomes			
	Pillai's Trace	F	df	p	Pillai's Trace	F	df	p	Pillai's Trace	F	df	p	Pillai's Trace	F	df	p
Group $\times$ condition $\times$ P.urgency	0.15	1.25	(8, 56)	0.29	0.01	0.28	(3, 63)	0.84	0.21	1.85	(8, 56)	0.09	0.02	0.36	(3, 62)	0.78
Two-way interaction																
Group $\times$ condition	0.18	1.52	(8, 57)	0.17	0.02	0.48	(3, 64)	0.70	0.16	1.37	(8, 57)	0.23	0.08	1.86	(3, 63)	0.15
Group $\times$ P.urgency	0.18	1.55	(8, 57)	0.16	0.07	1.68	(3, 64)	0.18	0.20	1.79	(8, 57)	0.10	0.06	1.35	(3, 63)	0.27
Condition $\times$ P.urgency	0.06	0.43	(8, 57)	0.90	0.08	1.88	(3, 64)	0.14	0.23	2.15	(8, 57)	<u>0.046</u>	0.06	1.25	(3, 63)	0.30
Main effects																
Group	0.19	1.67	(8, 57)	0.12	0.06	1.47	(3, 67)	0.23	0.22 [^]	2.05 [^]	(8, 57)	0.06	0.11	2.81	(3, 66)	<u>0.046</u>
Condition	0.06	0.45	(8, 57)	0.88	0.01	0.15	(3, 67)	0.93	0.22 [^]	1.96 [^]	(8, 57)	0.07	0.01	0.29	(3, 66)	0.84
P.urgency	0.14	1.14	(8, 57)	0.35	0.03	0.78	(3, 67)	0.51	0.26 [^]	2.49 [^]	(8, 57)	<u>0.02</u>	0.00	0.02	(3, 66)	1.00

Note: 2 (negative and positive urgency models) $\times$ 2 (cortical activity and behavioural outcomes) $\times$ 2 (Stroop and N-back tasks) separate MANOVAs were conducted; [^] values derived from two-way interaction model and not main effects model; Bonferroni adjusted $p < .006^*$; underlined $p < .05$.

MANOVA results. As several effects were significant at the conventional $p < .05$ cut-off, we report the findings based on this threshold next.

3.5.1. Stroop (SCWT)

3.5.1.1. ΔHbO_2 . Three-way interaction terms group $\times$ condition $\times$ negative/positive urgency were not significant, and thus we present findings from simpler models without the three-way interaction terms. In the two-way interaction model (negative urgency), significant effects were observed, including a group $\times$ negative urgency interaction [Pillai's Trace = 0.28, $F(8,57) = 2.78$, $p = .010$], a group $\times$ condition interaction [Pillai's Trace = 0.25, $F(8,57) = 2.33$, $p = .030$], and a main effect of group [Pillai's Trace = 0.3, $F(8,57) = 3.04$, $p = .006$]. Only the main effect of group was marginally significant at the Bonferroni-corrected threshold. No significant effects were observed for positive urgency model.

As MANOVA revealed significant effects of the negative urgency model at the standard threshold for cortical activity, we proceeded to conduct multiple linear regression analyses to locate the ROI(s) that harboured the significant effects. It was revealed that the group $\times$ negative urgency interaction ($b = -0.035$, $t = -2.41$, $p = .019$) and group main effect ($b = 0.07$, $t = 2.11$, $p = .039$) resided at the left OFC, and the group effect was also located at the right frontopolar cortex (FPC; $b = -0.039$, $t = -2.14$, $p = .037$). Group $\times$ condition interaction was marginally significant in the right FPC ($b = 0.017$, $t = 1.99$, $p = .05$) and was not analysed further.

As there was a significant interaction effect of group $\times$ negative urgency, we conducted post-hoc analysis of simple slopes to understand the nature of this interaction. It was revealed that higher negative urgency was significantly associated with reduced cortical activation in the left OFC ($b = -0.02$, $SE = 0.01$, $t = -2.4$, $p = .020$) in BD, but this relationship was not significant in HC ($b = 0.01$, $SE = 0.01$, $t = 1.16$, $p = .25$).

3.5.1.2. Behavioural outcomes. All interactions and main effects were not significant at the standard p -value threshold.

3.5.2. N-back

3.5.2.1. ΔHbO_2 . Like the results in Stroop, three way interaction terms were not significant. Therefore we present the results from simpler models without the three-way interaction terms. Based on the conventional $p < .05$ cut-off, in the negative urgency model, all interaction and main effects were not significant, whereas in the positive urgency model, analyses revealed a significant condition $\times$ positive urgency interaction [Pillai's Trace = 0.23, $F(8, 57) = 2.15$, $p = .046$] and positive urgency main effect [Pillai's Trace = 0.26, $F(8, 57) = 2.49$, $p = .022$]. All other effects were not significant.

To locate the effects, we conducted multiple regression analyses. These analyses revealed that the condition $\times$ positive urgency interaction effect ($b = 0.037$, $SE = 0.013$, $t = 2.811$, $p = .007$) and positive urgency main effect ($b = -0.03$, $SE = 0.013$, $t = -2.0$, $p = .049$) were observed at the right vlPFC. Condition $\times$ positive urgency interaction effect ($b = -0.014$, $SE = 0.006$, $t = -2.6$, $p = .011$) and positive urgency main effect ($b = 0.020$, $SE = 0.006$, $t = 3.244$, $p = .002$) resided at the left FPC.

Post-hoc simple slopes analysis of the condition $\times$ positive urgency interaction showed that a significant association between positive urgency and ΔHbO_2 activity in the right vlPFC ($b = 0.02$, $SE = 0.01$, $t = 2.24$, $p = .03$) was observed in the positive condition but not in the neutral condition ($b = -0.01$, $SE = 0.01$, $t = -1.48$, $p = .14$). On the other hand, a significant association between positive urgency and ΔHbO_2 activity in the left FPC was present in the neutral condition ($b = 0.02$, $SE = 0$, $t = 2.92$, $p \leq 0.01$) but absent in the positive condition ($b = 0.00$, $SE = 0.00$, $t = -0.73$, $p = .47$).

3.5.2.2. Behavioural outcomes. Here, we present findings based on the main effects model only, as three- and two-way interaction models were not significant. There was a significant main effect of group in both models [negative urgency model: Pillai's Trace = 0.13, $F(3, 67) = 3.32$, $p = .025$; positive urgency model: Pillai's Trace = 0.11, $F(3, 66) = 2.81$, $p = .046$]. Regression analyses indicate that the main effect of group was found for RT only (negative urgency model: $b = 0.13$, $SE = 0.05$, $t = 2.6$, $p = .011$; positive urgency model: $b = 0.14$, $SE = 0.05$, $t = 2.5$, $p = .013$). BD showed significantly greater reaction time latency (i.e., greater time contrast between 2-back and 1-back difficulty) than HCs. Neither negative nor positive urgency significantly predicted performance outcomes.

4. Discussion

The current study was unique in conjointly considering how bipolar disorder and urgency relate to cognitive control and working memory tasks and to cortical activation. Very few studies have considered neurocognitive facets of Urgency comparing those with and without bipolar disorder. We did so using two of the most commonly used tasks of EF, and multimodal behavioural and biological assessments. We begin by discussing the direct effects of diagnostic status and urgency. Then, we turn to our multivariate hypotheses and analyses, which considered the differential effects of positive vs. negative urgency, and how our cognitive and cortical outcomes were moderated by an experimental positive emotion induction and BD diagnosis.

Starting with main effects, we hypothesized that 1) the BD would perform more poorly and exhibit lower cortical activation than HC on EF tasks, and 2) urgency would predict poorer EF performance and lower PFC cortical activation. Behavioural data gave mixed support for hypothesis 1. That is, behavioural data suggests that the BD group did not perform significantly more poorly than HC in response errors, but this was partly due to prioritising accuracy over speed as a whole (poorer task inefficiencies in Stroop and N-back), which would have led to fewer hits and errors. For hypothesis 2, we observed small-to-medium correlations between positive urgency and EF performance (significant r ranging from 0.23 to 0.34).

We observed only limited support for these main effects in our PFC activation indices, in that only negative urgency was significantly (at standard unadjusted threshold) and inversely correlated with vlPFC activation (during Stroop task), to a small-to-medium degree. Additionally, a significant (marginally at Bonferroni adjusted threshold) group main effect was observed in two prefrontal regions during the Stroop task, with the bipolar group showing increased activation in the OFC and reduced activation in the right FPC. Contrary to expectations (Carroll et al., 2022; Johnson et al., 2020; Um et al., 2019), urgency was not significantly related to OFC activation.

Consistent with our third (exploratory) hypothesis, though, the main effects of urgency were qualified by significant interaction effects, with both diagnostic status and affect induction. We turn to those profiles next.

4.1. BD diagnostic status moderates urgency

Despite the null main effects for urgency with OFC activation, BD status appeared to be an important moderator, as this inverse relationship was only present among individuals with BD. This fits with previous findings suggesting that neurocognitive profiles of urgency generally are more commonly observed in clinical and diagnosed samples than nonclinical samples (Dekker and Johnson, 2018), with null findings reported in healthy undergraduates (Racine et al., 2024).

We considered mood state in BD as one probable explanation for this diagnostic interaction effect, as BD participants reported significantly lower mood levels at baseline (i.e., VAS pre-induction score). Among individuals with BD, negative urgency was positively correlated with depression levels to a moderate degree ($r = 0.35$, $p = .038$), negatively

correlated with left OFC activity to a similar degree ($r = -0.38$, $p = .023$), but not with right OFC activity ($r = -0.04$, ns) during cognitive control. These findings align with a previous report that found an inverse association between left OFC activity and depression severity among 10 euthymic and subsyndromal BD patients performing a variation of the Stroop task (Kronhaus et al., 2006).

4.2. Positive emotion context moderates emotion-based urgency

Although positive urgency showed small, nonsignificant effects on vIPFC and FPC activity ($r = |0.06|$ to $|0.20|$, ns), multiple MANOVA/MANCOVA analyses found that positive emotional context significantly moderated these relationships during the working memory task. The vIPFC, which is involved in regulating emotions and goal-directed behavior (Mitchell, 2011), had an inverse relationship with positive urgency during an emotionally neutral baseline. When induced with mild positive mood, however, individuals with high levels of positive urgency may require greater cognitive resources, reflected in heightened vIPFC processing, to regulate emotional impulse tendencies and counteract the detrimental effects of positive urgency on working memory performance. This pattern suggests that aberrant vIPFC activity may be relevant to urgency, particularly in experimental tasks with emotionally valenced stimuli or reward-based paradigms (Chase et al., 2017; Um et al., 2019). This finding helps clarify the mixed results in BD-relevant working memory research (Saldarini et al., 2022).

This study identified affective dispositional traits and context-dependent moderators, such as a positive emotional context, that could explain how PFC activity may be differentially linked to executive function. Indeed, the pattern of findings is striking given the modest mood induction effects, with significant changes in the cortical measures but not the VAS scores from pre- to post-induction.

4.3. Study limitations

It is important to consider several of the study's limitations. First, due to our limited sample size, our statistical models did not control for medication use, which could confound EF outcomes and cortical activity. Secondly, our exclusive focus on the PFC provides only a partial understanding of the neural mechanisms underlying EF performance and urgency. While using a small, non-invasive fNIRS neuroimaging device facilitated the inclusion of patients who might not participate if larger neuroimaging equipment were used, future research should aim to expand understanding by investigating subcortical regions and/or conducting network analyses. Such approaches could elucidate the functional interactions between different brain regions and their relationship to urgency. Third, our sample size was modest, which limits statistical power, particularly for detecting three-way interactions, and precluded control of some variables, such as medication use. Fourth, we used a short version of the UPPS, and the more limited reliability of the (lack of) Perseverance scale could contribute to the null findings for that dimension. Fifth, all MANOVA results were not significant at the Bonferroni-adjusted threshold, thus, our findings should be appraised as preliminary until they can be replicated in future research. Finally, due to ethical considerations to ensure safety and minimal risk to all participants, the paradigm elicited modest effects on mood, which could have limited the impact on overall findings.

4.4. Conclusion

Significant zero-order correlations were observed between various behavioural executive function (EF) outcomes and positive urgency, but not with negative urgency. Bipolar diagnosis and the positive affect context emerged as significant moderators in the relationships between EF neurocognitive outcomes and urgency subdimensions. Notably, the neurocognitive relationships with urgency subdomains, became significantly pronounced or changed in the presence of induced positive

emotion or among individuals with bipolar disorder (BD). These findings support the notion that urgency is indeed a clinically relevant phenotype, underscoring the need for greater exploration of its practical significance in the clinical domain of emotion-cognition neuroscience research.

CRedit authorship contribution statement

Wen Lin Teh: Writing – original draft, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Steve Yin Lam Lee:** Writing – review & editing, Resources. **Nisha Chandwani:** Writing – review & editing, Supervision, Resources. **Sheri L. Johnson:** Writing – review & editing. **Mythily Subramaniam:** Writing – review & editing, Supervision, Resources. **Roger Chun Man Ho:** Writing – review & editing, Supervision, Software, Resources.

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Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2025.119534>.

Data availability statement

Data for this study is available upon request. The restriction is imposed by our institutional and ethics committee. Data can be shared after a proposal is approved by the ethics committee. This has also been conveyed to our participants during the consent process. The data request can be sent to The Institutional Research Review Committee, Institute of Mental Health, Singapore; Email address: imhresearch@imh.com.sg.

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